Unified Curvature Certificates Across Learning, Aggregation, Attention, Arithmetic, and Symbolic Search

[Author redacted for review]

Abstract

The dynamics of the Quantum Arithmetic (QA) system correspond to multiplication by φ^2 in $\mathbb{Z}[\varphi]$, the ring of integers of $\mathbb{Q}(\sqrt{5})$: the map $T : (b, e) \mapsto (b{+}e,\ b{+}2e)$ equals Q^2, the square of the Fibonacci companion matrix, and acts as $\times\varphi^2$ on $b + e\varphi$. The Fibonacci quadratic norm $N(b + e\varphi) = b^2 + be - e^2$ is T-invariant, orbit lengths equal $\pi(m)/2$ (half the Pisano period), and the harmonic index H_{QA} admits a clean operator decomposition: $4H_{\mathrm{raw}} = \mathrm{tr}(\Lambda)$ where $\Lambda = \mathrm{diag}(\sigma_d, \sigma_{od})$ is a 2×2 cross-coupling operator whose degree-0 eigenvalue $\sigma_d = ba/(e^2 + d^2)$ tracks the projective Fibonacci angle and whose degree-1 eigenvalue $\sigma_{od} = ed/(b + a)$ collapses to $e/2$ via the Fibonacci identity $a + b = 2d$.

Building on this algebraic foundation, we introduce a certificate-normal-form framework for comparing one-step stability across heterogeneous computational systems. For any certified family whose local update writes as $p_{\mathrm{after}} = p_{\mathrm{before}} - \eta_{\mathrm{eff}} \cdot \mathrm{grad}$ with $\eta_{\mathrm{eff}} = \mathrm{lr} \cdot \mathrm{gain} \cdot H_{\mathrm{QA}}$, we define the universal curvature score

$$\kappa = 1 - |1 - \eta_{\mathrm{eff}}|.$$

We instantiate this across nine certified families: gradient-based learning [89], graph aggregation [93], attention layers [94], modular arithmetic dynamics [95], symbolic search [96], orbit curvature [97], spectral-gain extensions [98, 99], and gradient Lipschitz gain [101]. Three families ([98], [99], [101]) derive gain from native structural objects—the spectral norm of weight/score matrices and the gradient ℓ_2 norm—upgrading from consistency check to structural analysis. For QA orbit families, the finite enumerable orbit structure enables exact multi-step descent results: if $\kappa_{\min}(\mathcal{O}) > 0$, then $L_{t+L} \leq \rho(\mathcal{O}) \cdot L_t$ where $\rho(\mathcal{O}) = \prod_t (1 - \kappa_t)^2 < 1$ is computable exactly (scalar quadratic loss, Theorem 8.3); for any β-smooth μ-PL loss, $L(w_{t+L}) - L^* \leq \rho_{\mathrm{PL}}(\mathcal{O}) \cdot (L(w_t) - L^*) < L(w_t) - L^*$ (Theorem 8.5), with $\rho_{\mathrm{PL}}(\mathcal{O}) = \rho(\mathcal{O})$ when $\beta = \mu$. Empirically: across three independent architecture types (MLP gradient, GNN message-passing, single-head attention), mean κ correlates with final loss at $r \in \{-0.843, -0.810, -0.706\}$, all exceeding -0.70, with identical seven-substrate QA sweeps and same gain schedule. Normalizing $\eta_{\mathrm{eff}} = 1$ across substrates equalizes MLP convergence to loss std 1.9×10^{-5}, confirming that H_{QA} governs convergence exclusively through η_{eff}.

1 Introduction

The field of artificial intelligence is characterized by a proliferation of architectures. From multi-layer perceptrons to Graph Neural Networks (GNNs) and Transformers, each model family has its own stability considerations and hyperparameter conventions. Comparing the stability of a GNN's aggregation step with that of an attention head in a Transformer is not straightforward—each family uses different update semantics, different scale conventions, and different metadata. However, once each is mapped to a certified scalar local update form, comparison becomes possible.

This paper addresses this challenge by proposing a unified framework for certifying one-step stability across diverse computational architectures. Our primary contribution is the introduction of a curvature score κ, derived from a common mathematical substrate H_{QA}. This metric provides a single, interpretable value characterizing the single-step convergence behavior of a system's certified update rule, independent of domain-specific implementation details.

What this paper claims. The paper's claim is that nine heterogeneous architecture families admit the same machine-checkable one-step curvature certificate form, not that they share the same full dynamics. For three of these families ([98], [99], [101]) the gain is not a free witness but a derived structural invariant—the spectral norm of a native operator—making the certificate a structural analysis, not merely a consistency check. For QA orbit families specifically, Theorems 8.3 and 8.5 provide exact multi-step descent bounds for scalar quadratic and β-smooth μ-PL losses respectively. We do not assert general global convergence, multi-step equivalence across architectures, or semantic identity across architecture classes.

We demonstrate this across nine certified families:

1. **QALM Gradient [89]:** Standard gradient-based optimization.
2. **GNN Aggregation [93]:** The neighborhood aggregation step in GNNs.
3. **Attention Layer [94]:** The core mechanism in Transformer models.
4. **QARM Arithmetic [95]:** Operations within a modular arithmetic system.
5. **Symbolic Search [96]:** Heuristic-guided search algorithms.
6. **Orbit Curvature [97]:** QA orbit stability margin across a full finite orbit.
7. **GNN Spectral Gain [98]:** Gain derived from $\sigma_{\max}(W)$.
8. **Attention Spectral Gain [99]:** Gain derived from $\sigma_{\max}(QK^\top/\sqrt{d_k})$.
9. **Gradient Lipschitz Gain [101]:** Gain derived from $\|\mathbf{g}\|_2$, capped at 2.0.

Table 1: Nine certified families at a glance.

Family	Domain	Gain witness	Gain type	ID
QALM Gradient	Gradient optimization	`gain`	free scalar	[89]
GNN Aggregation	Graph neural networks	`agg_gain`	free scalar	[93]
Attention Layer	Transformer attention	`attn_gain`	free scalar	[94]
QARM Arithmetic	Modular arithmetic	`qarm_gain`	free scalar	[95]
Symbolic Search	Heuristic beam search	`sym_gain`	free scalar	[96]
Orbit Curvature	QA orbit stability	$\kappa_{\min}$	derived (orbit enum.)	[97]
GNN Spectral Gain	GNN weight geometry	$\sigma_{\max}(W)$	derived (power iter.)	[98]
Attention Spectral Gain	Attention geometry	$\sigma_{\max}(QK^\top/\sqrt{d_k})$	derived (power iter.)	[99]
Gradient Lipschitz Gain	Gradient descent	$\min(\|\mathbf{g}\|_2, 2)$	derived (L2 norm)	[101]

2 Mathematical Foundation

The H_{QA} substrate is not an ad-hoc formula. It arises from the arithmetic of $\mathbb{Q}(\sqrt{5})$ and from the projective dynamics of the Fibonacci recurrence. We summarise the structure here; formal proofs appear in Section 12.

The ring $\mathbb{Z}[\varphi]$ and the QA map. The golden ratio $\varphi = (\sqrt{5}+1)/2$ satisfies $\varphi^2 = \varphi+1$. The ring of integers $\mathbb{Z}[\varphi] = \{b+e\varphi : b, e \in \mathbb{Z}\}$ carries the Fibonacci quadratic norm $N(b+e\varphi) = b^2+be-e^2$. The QA map $T : (b, e) \mapsto (d, a)$ with $d = b+e$, $a = b+2e$ (before modular reduction) corresponds to multiplication by φ^2:

$$\varphi^2(b+e\varphi) = (b+e) + (b+2e)\varphi = d + a\varphi.$$

The matrix of T is $Q^2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, the square of the Fibonacci companion matrix $Q = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

Norm invariance and orbit structure. Since $N(\varphi^2) = 1$, the norm $N(b+e\varphi) = b^2+be-e^2$ is T-invariant. Modulo m, the orbit length equals $\pi(m)/2$—half the Pisano period—because the order of Q^2 in $GL_2(\mathbb{Z}/m\mathbb{Z})$ is $\pi(m)/2$ for all $m \geq 3$ (Propositions 12.1 and 12.2). For $m = 9$: $\pi(9) = 24$, orbit length $= 12$, 72 starting pairs ("Cosmos" group). The modular value $N(b + e\varphi) \bmod m$ classifies orbits: for $m = 9$, $3 \nmid N$ gives cosmos (length 12), $3^2 \mid N$ gives satellite (length 4 or 1), $3^4 \mid N$ gives the singularity (length 1).

Projective Fibonacci flow. The projective ratio $z = e/(b + e)$ evolves as the Möbius map $z \mapsto (z + 1)/(z + 2)$ under T, with unique fixed point $z^* = 1/\varphi \approx 0.618$. Every pre-modular orbit converges projectively to z^*.

H_{QA} as a cross-coupling trace. The step matrix $M = \left(\begin{smallmatrix} b & e \\ d & a \end{smallmatrix}\right)$ (rows = state before and after T) satisfies $\det(M) = N(b + e\varphi)$ (pre-modular). The harmonic index decomposes as:

$$4H_{\mathrm{raw}} = \underbrace{\frac{ba}{e^2+d^2}}_{\sigma_d,\ \text{degree-0}} + \underbrace{\frac{ed}{b+a}}_{\sigma_{od},\ \text{degree-1}} = \mathrm{tr}(\Lambda), \qquad \Lambda = \mathrm{diag}(\sigma_d,\, \sigma_{od}).$$

The degree-0 component $\sigma_d = \cos(2\arctan(z))$ tracks the projective Fibonacci angle and is scale-invariant; the degree-1 component $\sigma_{od} = e/2$ is fixed by the Fibonacci identity $a + b = 2d$ and carries the amplitude. The factor $1/4$ is the normalised-trace scaling: $H_{\mathrm{raw}} = \mathrm{tr}(\Lambda)/4 = R(\Lambda, (1, 1)^\top/\sqrt{2})/2$ (Proposition 12.4). Properties (i) degree-(0,1) homogeneity, (ii) transpose symmetry $\sigma(M) = \sigma(M^\top)$, and (iii) projective Fibonacci convergence characterise $4H_{\mathrm{raw}}$ as the natural cross-coupling trace of the QA step matrix (Section 12).

3 The H_{QA} Substrate

Section 2 established the algebraic origin of H_{QA}. We now give the formal computational definition used by all validators.

3.1 Formal Definition

Given four positive integers $a, b, d, e \geq 1$, define:

$$G = e \cdot e + d \cdot d, \qquad F = b \cdot a.$$

From these, compute a raw harmonic ratio H_{raw} with a small $\varepsilon = 10^{-12}$:

$$H_{\text{raw}} = 0.25 \cdot \left(\frac{F}{G + \varepsilon} + \frac{e \cdot d}{a + b + \varepsilon} \right).$$

3.2 Boundedness Property

The saturating function

$$H_{\text{QA}} = \frac{|H_{\text{raw}}|}{1 + |H_{\text{raw}}|}$$

maps H_{raw} to $[0, 1)$. This ensures H_{QA} is always non-negative and strictly less than 1, a critical prerequisite for the stability analysis of the universal κ formula.

3.3 Geometric Interpretation

As derived in Section 2, $4H_{\text{raw}} = \text{tr}(\Lambda)$ where $\Lambda = \text{diag}(\sigma_d, \sigma_{od})$ is the cross-coupling operator of the QA step matrix. The degree-0 eigenvalue $\sigma_d = ba/(e^2 + d^2) = \cos(2\theta)$ where $\theta = \arctan(e/d)$; the degree-1 eigenvalue $\sigma_{od} = ed/(b+a) = e/2$ carries the state amplitude. Both terms are algebraically determined by the $\mathbb{Q}(\sqrt{5})$ structure; the normalisation constant $1/4$ carries a residual derivation gap (Section 12).

4 The Universal κ Formula

4.1 Derivation

Consider a general iterative process where a parameter p is updated at each step. We define an *effective learning rate* modulated by the system's substrate:

$$\eta_{\text{eff}} = \text{lr} \cdot \text{gain} \cdot H_{\text{QA}}.$$

Here lr is a global base learning rate, H_{QA} is the Harmonic Index of the system's current state, and gain is a scalar witness specific to the architecture class. The certified update rule becomes:

$$p_{\text{after}} = p_{\text{before}} - \eta_{\text{eff}} \cdot \text{grad}.$$

We define the curvature score:

$$\kappa = 1 - |1 - \eta_{\text{eff}}| = 1 - |1 - \text{lr} \cdot \text{gain} \cdot H_{\text{QA}}|.$$

Interpretation. The quantity $\kappa \in (-\infty, 1]$ is positive exactly when $\eta_{\text{eff}} \in (0, 2)$ and is maximised at $\eta_{\text{eff}} = 1$. It measures proximity to the centre of the admissible one-step interval, not the absolute magnitude of the step.

4.2 Stability Theorem

Theorem 4.1 (One-Step Stability)**.** *The single-step update is stable ($\kappa > 0$) if and only if* $\eta_{\text{eff}} \in (0, 2)$.

Proof. $\kappa > 0 \Leftrightarrow |1 - \eta_{\text{eff}}| < 1 \Leftrightarrow 0 < \eta_{\text{eff}} < 2$. A practical sufficient condition is gain $\in (0, 2]$ and lr < 1; since $H_{\text{QA}} < 1$, this gives $\eta_{\text{eff}} < 2$. □

5 Nine Architecture Classes

We instantiate the framework across nine certified families. Cross-row differences in Table 2 arise from different substrate and gain inputs, not architecture identity. Families [98], [99], and [101] derive gain from a native operator; the remainder supply a free scalar.

5.1 QALM Gradient [89]

Gain: `gain`. **Description:** The `gain` parameter acts as a scalar on the learning rate, modulated by H_{QA} of the current state. **Example:** Substrate $(b, e, d, a) = (1, 2, 3, 5)$, $H_{\text{QA}} \approx 0.2571$.

5.2 GNN Aggregation [93]

Gain: `agg_gain`. **Metadata:** `n_nodes`, `n_edges`. **Description:** Controls the strength of the aggregated message updating a node's representation. Structural metadata provides auditable graph context but does not enter the curvature computation.

5.3 Attention Layer [94]

Gain: `attn_gain`. **Metadata:** `n_heads`, `d_model`, `seq_len`. **Description:** Scales the output of the attention head.

5.4 QARM Arithmetic [95]

Gain: `qarm_gain`. **Metadata:** `modulus`, `orbit_size`, `generator`.

5.5 Symbolic Search [96]

Gain: `sym_gain`. **Metadata:** `beam_width`, `search_depth`, `rule_count`. **Example:** Substrate $(3, 5, 8, 13)$, $H_{\text{QA}} \approx 0.4235$, lr $= 0.01$, sym_gain $= 1.1$: $\kappa \approx 0.00466$.

5.6 Orbit Curvature [97]

Gain: $\kappa_{\min}$ derived by full orbit enumeration. **Description:** Rather than a single-state snapshot, this family certifies the tightest stability bottleneck across an entire QA orbit. The validator enumerates the complete orbit under T modulo m, computes H_{QA} at every state, and reports $\kappa_{\min} = \min_t \kappa_t$.

Orbit length: $|\mathcal{O}| = \pi(m)/2$ (Proposition 12.1). For $m = 9$: $\pi(9) = 24$, orbit length 12, 72 starting pairs. **Example:** Orbit start $(b_0, e_0) = (1, 2)$, modulus 9, lr $= 0.5$, gain $= 1.0$: $\kappa_{\min} \approx 0.1077$ (bottleneck at $(8, 1, 9, 1)$), orbit length 12.

5.7 GNN Spectral Gain [98] and Attention Spectral Gain [99]

Gain (GNN): $\sigma_{\max}(W)$ via power iteration on $W^\top W$. **Gain (Attention):** $\sigma_{\max}(QK^\top/\sqrt{d_k})$ via power iteration. Gate B computes the gain internally and rejects any certificate where the claimed value disagrees.

GNN example: $W = \left(\begin{smallmatrix} 0.8 & 0.2 \\ 0.1 & 0.6 \end{smallmatrix}\right)$, substrate $(3, 5, 8, 13)$, $H_{\mathrm{QA}} \approx 0.4235$, $\sigma_{\max}(W) \approx 0.8821$, $\kappa \approx 0.00374$.

Attention example: $Q = \left(\begin{smallmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{smallmatrix}\right)$, $K = \left(\begin{smallmatrix} 0.5 & 0.3 \\ 0.2 & 0.8 \end{smallmatrix}\right)$, $d_k = 2$, $\sigma_{\max}(QK^\top/\sqrt{2}) \approx 0.6603$, $\kappa \approx 0.00280$.

5.8 Gradient Lipschitz Gain [101]

Gain: $\min(\|\mathbf{g}\|_2, 2.0)$ derived from the submitted gradient. **Description:** The L2 norm of the gradient is the natural local Lipschitz constant of the update step. Gate B recomputes the norm and rejects disagreements. **Example:** $\mathbf{g} = [0.3, 0.4]$, $\|\mathbf{g}\|_2 = 0.5$, substrate $(3, 5, 8, 13)$, $H_{\mathrm{QA}} \approx 0.4235$, $\mathrm{lr} = 0.01$, $\kappa \approx 0.00212$.

6 The Three-Gate Certificate Pattern

Any artifact claiming compliance must pass a three-gate validation:

- **Gate A – Substrate Integrity:** The validator recomputes H_{QA} from raw parameters (a, b, d, e) and compares to the claimed value. Failure mode: `H_QA_MISMATCH`.
- **Gate B – Update Rule Integrity:** The validator checks gain $\in (0, 2]$, recomputes η_{eff}, and verifies the claimed p_{after}. Failure modes: `GAIN_OUT_OF_RANGE`, `UPDATE_RULE_MISMATCH`.
- **Gate C – Curvature Integrity:** Using verified inputs, the validator recomputes κ and compares. Failure mode: `KAPPA_MISMATCH`.

The failure types are disjoint: a certificate fails at exactly one gate, and the `invariant_diff` field identifies the specific numerical discrepancy.

7 Comparative Certificate Instantiations

Table 2 shows that once a family is in the certified normal form, the resulting κ depends only on $(\mathrm{lr}, \mathrm{gain}, H_{\mathrm{QA}})$, not on the architectural label. We use $\mathrm{lr} = 0.01$, $\mathrm{gain} = 1.1$.

For a given substrate, κ is identical across all five families despite their different purposes. This demonstrates that κ is sensitive to the underlying scalar inputs while being independent of architectural specifics.

8 Theorems

Theorem 8.1 (Unified Curvature Normal Form)**.** *Let a certified family provide a one-step update*

$$p_{\mathrm{after}} = p_{\mathrm{before}} - \eta_{\mathrm{eff}} \cdot \mathrm{grad}, \qquad \eta_{\mathrm{eff}} = \mathrm{lr} \cdot \mathrm{gain} \cdot H_{\mathrm{QA}},$$

where H_{QA} is computed from a four-parameter substrate (a, b, d, e) by the definitions in Section 3, $\mathrm{lr} \geq 0$, $\mathrm{gain} \geq 0$. *Then the associated one-step curvature score $\kappa = 1 - |1 - \eta_{\mathrm{eff}}|$ is positive if and*

Table 2: Cross-family comparison at two substrates (lr = 0.01, gain = 1.1).

Family	ID	Gain name	H_{QA}	gain	κ
QALM Gradient	[89]	`gain`	0.2571	1.1	0.00283
GNN Aggregation	[93]	`agg_gain`	0.2571	1.1	0.00283
Attention Layer	[94]	`attn_gain`	0.2571	1.1	0.00283
QARM Arithmetic	[95]	`qarm_gain`	0.2571	1.1	0.00283
Symbolic Search	[96]	`sym_gain`	0.2571	1.1	0.00283
QALM Gradient	[89]	`gain`	0.4235	1.1	0.00466
GNN Aggregation	[93]	`agg_gain`	0.4235	1.1	0.00466
Attention Layer	[94]	`attn_gain`	0.4235	1.1	0.00466
QARM Arithmetic	[95]	`qarm_gain`	0.4235	1.1	0.00466
Symbolic Search	[96]	`sym_gain`	0.4235	1.1	0.00466

only if $0 < \eta_{\text{eff}} < 2$. *In particular, any certified family whose validator enforces this normal form inherits the same scalar one-step stability criterion, regardless of domain-specific interpretation of the gain witness.*

Remark 8.2. This theorem is a statement about a shared certified local update form. It does not assert global convergence or semantic identity across architecture classes. For QA orbit families, Theorems 8.3 and 8.5 extend this to exact multi-step bounds.

8.1 Finite-Orbit Descent Theorem (Quadratic Loss)

The finite structure of the QA orbit enables a multi-step descent result.

Theorem 8.3 (Finite-Orbit Descent, Quadratic Loss)**.** *Let* $L(w) = \frac{1}{2}w^2$ *be a scalar quadratic loss. Let* $\mathcal{O} = \{s_0, \ldots, s_{L-1}\}$ *be a QA cosmos orbit of length* $L = \pi(m)/2$ *with per-step effective rates* $\eta_{\text{eff}}^{(t)} = \text{lr} \cdot \text{gain} \cdot H_{\text{QA}}(s_t)$ *and curvature scores* $\kappa_t = 1 - |1 - \eta_{\text{eff}}^{(t)}|$. *Define the orbit contraction factor*

$$\rho(\mathcal{O}) := \prod_{t=0}^{L-1} (1 - \kappa_t)^2.$$

If $\kappa_{\min}(\mathcal{O}) := \min_t \kappa_t > 0$, *then:*

(i) $L_{t+L}(w) \leq \rho(\mathcal{O}) \cdot L_t(w)$ *for every initial* w*;*

(ii) $\rho(\mathcal{O}) \leq (1 - \kappa_{\min})^{2L} < 1$*;*

(iii) $\rho(\mathcal{O})$ *is a computable exact quantity from the orbit alone, with no stochastic approximation.*

Proof. For the scalar quadratic, $w_{t+1} = (1-\eta_{\text{eff}}^{(t)})w_t$, so $L_{t+1} = (1-\eta_{\text{eff}}^{(t)})^2 L_t$. Since $|1-\eta_{\text{eff}}^{(t)}| = 1-\kappa_t$, we have $L_{t+1} = (1 - \kappa_t)^2 L_t$. Composing over L steps: $L_{t+L} = \rho(\mathcal{O}) \cdot L_t$. For (ii): $(1 - \kappa_t)^2 \leq (1 - \kappa_{\min})^2$ for all t, so $\rho \leq (1 - \kappa_{\min})^{2L} < 1$ since $\kappa_{\min} > 0$. For (iii): each κ_t is computable from s_t via Section 3, and $L = \pi(m)/2$ is finite by Proposition 12.1. □

Corollary 8.4 (Multi-orbit convergence)**.** *After* k *full orbits,* $L_{kL} \leq \rho(\mathcal{O})^k \cdot L_0 \to 0$ *geometrically with rate* $\log(1/\rho(\mathcal{O}))$ *per orbit.*

Key identity. The per-step loss contraction factor is $(1-\kappa_t)^2$: higher κ directly and exactly governs loss decrease. This gives the first-principles explanation of the empirical correlation $r(\text{mean}\,\kappa, \text{final loss}) = -0.843$ reported in Section 11.

Numerical example ($m = 9$, lr $= 0.5$, gain $= 1$). The mod-9 cosmos orbit has $L = 12$ states. Numerical evaluation yields $\kappa_{\min} = 0.1077$ (bottleneck at state $(8, 1, 9, 1)$) and

$$\rho(\mathcal{O}) = \prod_{t=0}^{11}(1-\kappa_t)^2 = 0.001582,$$

so every full orbit reduces loss by a factor of 632. The orbit-invariant $\rho(\mathcal{O})$ is the same for all 72 starting pairs in the cosmos group (they all traverse the same 12-cycle). After 10 orbits: $L_{10L} \leq (0.001582)^{10} \cdot L_0 \approx 10^{-28} \cdot L_0$.

8.2 Extension to PL-Conditioned Losses

The scalar quadratic is a special case ($\beta = \mu$) of a broader class.

Theorem 8.5 (Finite-Orbit Descent, PL Loss)**.** *Let $L : \mathbb{R}^d \to \mathbb{R}$ be β-smooth and satisfy the Polyak-Łojasiewicz (PL) condition with constant $\mu > 0$: $\|\nabla L(w)\|^2 \geq 2\mu(L(w) - L^*)$ for all w, where $L^* = \inf_w L(w)$. Let $\mathcal{O}$ be a QA cosmos orbit with $\kappa_{\min}(\mathcal{O}) > 0$ and $\eta_{\text{eff}}^{(t)} < 2/\beta$ for all t. Define*

$$\rho_{\text{PL}}(\mathcal{O}) := \prod_{t=0}^{L-1}\left(1 - 2\mu\,\eta_{\text{eff}}^{(t)}\left(1 - \tfrac{\beta}{2}\eta_{\text{eff}}^{(t)}\right)\right).$$

Then $0 \leq \rho_{\text{PL}}(\mathcal{O}) < 1$ and $L(w_{t+L}) - L^* \leq \rho_{\text{PL}}(\mathcal{O}) \cdot (L(w_t) - L^*)$.

Proof. At each step, the β-smooth descent lemma gives $L(w_{t+1}) \leq L(w_t) - \eta_{\text{eff}}^{(t)}(1 - \frac{\beta}{2}\eta_{\text{eff}}^{(t)})\|\nabla L(w_t)\|^2$. Applying the PL condition: $L(w_{t+1}) - L^* \leq \big(1 - 2\mu\eta_{\text{eff}}^{(t)}(1 - \frac{\beta}{2}\eta_{\text{eff}}^{(t)})\big)(L(w_t) - L^*)$. Since $\eta_{\text{eff}}^{(t)} \in (0, 2/\beta)$, each factor is in $[0, 1)$. Composing over L steps gives $\rho_{\text{PL}}(\mathcal{O}) < 1$. □

Reduction to Theorem 8.3. For quadratic loss with Hessian λI (so $\beta = \mu = \lambda$), the per-step factor becomes $1 - 2\lambda\eta(1 - \frac{\lambda}{2}\eta) = (1 - \lambda\eta)^2 = (1 - \kappa_t)^2$, so $\rho_{\text{PL}}(\mathcal{O}) = \rho(\mathcal{O})$. This identity holds exactly for all $\eta_{\text{eff}} \in (0, 2)$ (verified numerically). Theorem 8.3 is the canonical case of Theorem 8.5.

Condition number dependence. For the mod-9 cosmos orbit at lr $= 0.5$: $\rho_{\text{PL}}(\mathcal{O}) = 0.0016$ (condition 1), 0.61 (condition 10), 0.95 (condition 100). The orbit still guarantees descent for any finite condition number, with rate degrading gracefully.

8.3 Extension to Łojasiewicz Losses

The results of §§8–8.2 are exact for quadratic and β-smooth μ-PL losses. We now prove an extension to the broader Łojasiewicz class ($\alpha \in (0, 1)$), under an explicit trajectory hypothesis. The proof uses a change of variable $\varphi_t = V_t^{1-\alpha}$ that converts the nonlinear per-step bound into an additive orbit-window decrement.

Theorem 8.6 (Finite-Orbit Descent, Łojasiewicz Loss)**.** *Let $L : \mathbb{R}^d \to \mathbb{R}$ be β-smooth and satisfy the Łojasiewicz condition*

$$\|\nabla L(w)\|^2 \geq 2\mu(L(w) - L^*)^\alpha \quad \forall\, w \in S_0 := \{w : L(w) \leq L(w_t)\}$$

for some $\mu > 0$, $\alpha \in (0,1)$, where $L^* = \inf_w L(w)$. Let $\mathcal{O}$ be a QA cosmos orbit with $\eta_{\text{eff}}^{(s)} \in (0, 2/\beta)$ for all s, and suppose

$$\nabla L(w_s) \neq 0 \quad \textit{for } s = t, \ldots, t+L-1 \qquad \textit{(H-crit)}$$

Define $V_t = L(w_t) - L^*$, $\varphi_t = V_t^{1-\alpha}$, and

$$C(\mathcal{O}) := \sum_{s=0}^{L-1} 2\mu\, \eta_{\text{eff}}^{(s)} \Big(1 - \tfrac{\beta}{2}\eta_{\text{eff}}^{(s)}\Big) > 0.$$

Then either (A) $V_{t+L} = 0$ *(convergence at the orbit endpoint), or* (B) $\varphi_{t+L} \leq \varphi_t - (1-\alpha)\, C(\mathcal{O})$.

Proof. (H-crit) implies $V_s > 0$ for $s \in \{t, \ldots, t+L-1\}$ (otherwise w_s is a minimiser, so $\nabla L(w_s) = 0$, contradicting (H-crit)). The β-smooth descent lemma and Łojasiewicz condition give $V_{s+1} \leq V_s - c_s V_s^{\alpha}$ where $c_s = 2\mu\eta_{\text{eff}}^{(s)}(1 - \frac{\beta}{2}\eta_{\text{eff}}^{(s)}) > 0$. All iterates remain in S_0 by induction ($L(w_{s+1}) \leq L(w_s)$), so the Łojasiewicz condition applies throughout.

Split on V_{t+L}. If $V_{t+L} = 0$: conclusion (A). If $V_{t+L} > 0$: at every step $s \in \{t, \ldots, t+L-1\}$, the overshoot case ($V_{s+1} = 0$) is ruled out — by (H-crit) for $s < t+L-1$, and by $V_{t+L} > 0$ for the final step. Hence at each step, the concavity of $f(x) = x^{1-\alpha}$ (since $f'' < 0$) gives via the tangent-line bound:

$$\varphi_{s+1} = V_{s+1}^{1-\alpha} \leq (V_s - c_s V_s^{\alpha})^{1-\alpha} \leq V_s^{1-\alpha} - (1-\alpha)c_s = \varphi_s - (1-\alpha)c_s.$$

Telescoping over L steps yields conclusion (B). □

Convergence rate. If $V_t > 0$ for all complete orbits, convergence occurs within $\lceil \varphi_0 / [(1-\alpha)C(\mathcal{O})] \rceil$ orbits. The orbit contributes through $C(\mathcal{O})$: higher $\kappa_{\min}$ raises $\eta_{\text{eff}}^{(s)}$, raises each c_s, and accelerates convergence. $\kappa_{\min}$ remains the central orbit scalar.

Relation to Theorem 8.5. Theorem 8.6 is a *sublinear analogue* of Theorem 8.5, not a reduction to it. The PL result gives multiplicative contraction in V-coordinates; this theorem gives additive decay in φ-coordinates. The two are not equivalent at $\alpha = 1$: the PL result is strictly stronger.

Genericity of (H-crit). If $\text{crit}(L) = \{w : \nabla L(w) = 0\}$ has Lebesgue measure zero (in particular if L is a Morse function or polynomial), then for Lebesgue-almost-every starting point w_0, hypothesis (H-crit) holds. The bad set $\bigcup_{s=0}^{L-1} F_s^{-1}(\text{crit}(L))$ is a finite union of measure-zero sets (by the Federer area formula applied to the gradient-flow maps F_s), hence has measure zero.

Cert family [102]. Theorem 8.6 is operationalized by cert family [102] (`QA_LOJASIEWICZ_ORBIT_CERT.v1`). Gate 2D recomputes $C(\mathcal{O}) = \sum_s 2\mu\eta_{\text{eff}}^{(s)}(1 - \frac{\beta}{2}\eta_{\text{eff}}^{(s)})$ from the orbit η_{eff} sequence and verifies the claimed value to within 10^{-9}. H-crit is an explicit witnessed boolean in v1; the intrinsic H-crit-free extension remains open.

Open problem. A fully intrinsic version — depending only on the orbit $\mathcal{O}$ and loss class, with no condition on w_0 — remains open. Closure would require either a Łojasiewicz argument that structurally avoids saddle-point trajectories, or a Morse-theoretic bound on QA orbit windows.

9 Limitations

For general nonconvex losses, this framework certifies only a local one-step normal form and does not prove multi-step convergence, generalization, or robustness under distribution shift. Theorems 8.3, 8.5, and 8.6 give exact or bounded multi-step guarantees for scalar quadratic, β-smooth μ-PL, and Łojasiewicz ($\alpha < 1$) losses respectively; the Łojasiewicz result (Theorem 8.6) requires an explicit trajectory hypothesis (H-crit) whose generic validity is established as a separate proposition. A fully intrinsic nonconvex extension remains open.

In families [89]–[96], the gain is supplied as a free scalar witness; the metadata fields (`n_nodes`, `d_model`, `orbit_size`, etc.) are recorded for auditing but do not constrain that scalar. Accordingly, those families should be read as reusable machine-checkable local certification patterns. Families [98], [99], and [101] close this gap: gain is derived from $\sigma_{\max}(W)$, $\sigma_{\max}(QK^\top/\sqrt{d_k})$, and $\|\mathbf{g}\|_2$ respectively, and cannot be freely adjusted. Extending derived-gain derivations to the remaining families (QARM, symbolic search) is left to future work.

10 Conclusion

This paper introduced a certificate-normal-form framework grounded in the arithmetic of $\mathbb{Q}(\sqrt{5})$: the QA map $T = Q^2$ acts as multiplication by φ^2 in $\mathbb{Z}[\varphi]$, and the harmonic index H_{QA} decomposes as the normalised trace of a natural cross-coupling operator on the QA step matrix. Building on this foundation, we derived a universal curvature score κ and certified its one-step stability form across nine architecture families.

For QA orbit families, Theorems 8.3, 8.5, and 8.6 provide multi-step descent guarantees across three loss classes: if $\kappa_{\min}(\mathcal{O}) > 0$, one full orbit reduces scalar quadratic loss by exactly computable factor $\rho(\mathcal{O}) < 1$ (Theorem 8.3); for any β-smooth μ-PL loss, $\rho_{\mathrm{PL}}(\mathcal{O}) < 1$ (§8.2); and for any β-smooth Łojasiewicz loss with $\alpha \in (0, 1)$, either convergence occurs within the window or $\varphi_{t+L} \leq \varphi_t - (1-\alpha)C(\mathcal{O})$ where $C(\mathcal{O}) > 0$ is orbit-computable (§8.3, under H-crit). For the mod-9 cosmos orbit at $\mathrm{lr} = 0.5$, $\rho(\mathcal{O}) = 0.001582$ (factor-632 per orbit). Cert family [102] operationalizes Theorem 8.6.

Future work: (1) derive an intrinsic H-crit-free version of Theorem 8.6; (2) derive gain from native objects in QARM and symbolic search families; (3) explore κ as an active regularisation term in training; (4) develop a standalone mathematical note on the $\mathbb{Z}[\varphi]$/Half-Pisano/norm-invariant structure.

11 Empirical Validation

We validate the κ framework with two controlled experiments on synthetic binary classification (800 samples, 20 features, two Gaussian classes). The model is a two-layer MLP (20 $\to$ 32 ReLU$\to$ 1 sigmoid) trained with binary cross-entropy, pure NumPy, no GPU.

Setup.

$$p_{\mathrm{after}} = p_{\mathrm{before}} - \mathrm{lr} \cdot \mathrm{gain} \cdot H_{\mathrm{QA}} \cdot \nabla\mathcal{L}, \qquad \kappa = 1 - |1 - \mathrm{lr} \cdot \mathrm{gain} \cdot H_{\mathrm{QA}}|.$$

Experiment 1 – H_{QA} sweep. Seven substrates spanning $H_{\mathrm{QA}} \in [0.20, 0.71]$ (mod-9, gain $= \min(\|\mathbf{g}\|_2, 2)$, $\mathrm{lr} = 0.1$, 300 epochs) plus plain-SGD baseline.

Table 3: Experiment 1: H_QA sweep results.

Substrate	H_{QA}	mean κ	Final loss
(2,8)	0.201	0.0035	0.0434
(4,7)	0.305	0.0043	0.0395
(1,4)	0.356	0.0052	0.0407
(5,3)	0.471	0.0056	0.0289
(9,8)	0.529	0.0065	0.0279
(3,5)	0.594	0.0068	0.0254
(1,5)	0.715	0.0077	0.0231

The ordering is monotone: higher $H_{\mathrm{QA}} \to$ higher $\kappa \to$ lower final loss. Pearson correlations: $r(\text{mean}\,\kappa, \text{final loss}) = -0.843$, $r(H_{\mathrm{QA}}, \text{final accuracy}) = +0.769$. Plain SGD achieves lower absolute loss because gain $= \|\mathbf{g}\|_2 \ll 1$ in the QA conditions shrinks the effective step by 20–150×; within the QA family the κ ordering is preserved and strongly predictive.

Experiment 2A – η_{eff} sweep. Fixed substrate $(9, 8)$ ($H_{\mathrm{QA}} = 0.529$), gain $= 1$, lr $= \eta_{\mathrm{eff}}/H_{\mathrm{QA}}$ exact.

Table 4: Experiment 2A: η_{eff} sweep (substrate (9,8)).

η_{eff}	κ	Final loss
0.05	0.05	1.77×10^{-3}
0.25	0.25	2.32×10^{-4}
0.50	0.50	1.05×10^{-4}
1.00	1.00	4.6×10^{-5}
1.50	0.50	1.1×10^{-5}
1.90	0.10	1.9×10^{-5}

Loss decreases monotonically from $\eta_{\mathrm{eff}} = 0.05$ to 1.50, then stabilises. Pearson $r(\kappa, \text{final loss}) = -0.53$.

Experiment 2B – substrate equalization. Setting lr $= 1/H_{\mathrm{QA}}$ for each of seven substrates forces $\kappa = 1$ uniformly. All seven converge in epoch 1; loss standard deviation 1.9×10^{-5} (range $[7 \times 10^{-6}, 7 \times 10^{-5}]$). This confirms that H_{QA} governs convergence only through η_{eff}; once η_{eff} is equalized, substrate identity has negligible effect.

Summary. Three claims supported jointly: (i) within the QA update family, mean κ is a strong predictor of convergence quality ($r = -0.843$); (ii) the critical design parameter is η_{eff}, with κ characterizing proximity to the stability boundary; (iii) once η_{eff} is equalized, convergence is equalized.

Experiment 3 – GNN message passing [93]. We replicate the sweep on a 2-layer GCN trained for node classification on a 200-node stochastic block model graph (2 communities, $p_{\mathrm{in}} = 0.30$, $p_{\mathrm{out}} = 0.05$, 16 node features). The message-passing update is $H^{(l+1)} = \mathrm{ReLU}(\tilde{A}H^{(l)}W^{(l)})$ where $\tilde{A}$ is the symmetric normalised adjacency with self-loops. The κ-modulated rule applies to both weight matrices $W^{(1)}, W^{(2)}$; gain is the ℓ_2-norm of the concatenated gradient. Same 7 substrates,

lr = 0.05, 300 epochs. Result: $r(\text{mean}\,\kappa, \text{final loss}) = -0.810$ — above the -0.70 threshold, confirming that κ predicts convergence in message-passing architectures as strongly as in the MLP baseline.

Experiment 4 – Self-attention layer [94]. A single-head self-attention classifier (128 sequences $\times$ 8 tokens $\times$ 16 dims, $d_k = 16$, mean-pool readout, binary BCE) is trained under the same substrate sweep (lr = 0.10, 300 epochs). Gain is the ℓ_2-norm of all five weight-matrix gradients $(W_Q, W_K, W_V, W_O, W_{\text{cls}})$. Softmax in the attention mechanism distributes gradients across tokens, systematically reducing per-parameter gradient norms and hence compressing the effective κ range. Result: $r(\text{mean}\,\kappa, \text{final loss}) = -0.706$, above threshold and exhibiting the same qualitative pattern: higher $H_{\text{QA}} \to$ lower final loss across the seven substrates.

Table 5: Cross-architecture κ predictiveness: $r(\text{mean}\,\kappa, \text{final loss})$ in three independent experiments.

Architecture	Family	Base lr	$r(\kappa, \text{loss})$
MLP gradient [89]	Exp. 1	0.10	-0.843
GNN aggregation [93]	Exp. 3	0.05	-0.810
Attention layer [94]	Exp. 4	0.10	-0.706

All three experiments use identical QA substrates, the same gain schedule, and the same 7-condition sweep. The consistent negative correlation across architectures with substantially different gradient structures (dense MLP, graph-normalised aggregation, softmax-attenuated) supports the universality claim of the κ framework.

12 Appendix: Mathematical Proofs

Proposition 12.1 (Half-Pisano Orbit Length)**.** *For every $m \geq 3$, the QA update map $T : (b, e) \mapsto (b{+}e \bmod {}^{*}_{m},\ b{+}2e \bmod {}^{*}_{m})$ satisfies $T = Q^2$ and has maximal orbit length $\pi(m)/2$, where $\pi(m)$ is even for all $m \geq 3$. A state $(b_0, e_0) \in \{1, \ldots, m\}^2$ is* primitive *if its orbit under T attains this maximal length.*

Proof sketch. Step 1 – Matrix identity.

$$T \begin{pmatrix} b \\ e \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} b \\ e \end{pmatrix} = Q^2 \begin{pmatrix} b \\ e \end{pmatrix} \pmod{m}.$$

Step 2 – Pisano period is even for $m \geq 3$. Wall [9] shows that for $m \geq 3$, $\pi(m) \equiv 0 \pmod 2$.

Step 3 – Order of Q^2. $\text{ord}(Q^2) = \pi(m)/\gcd(2, \pi(m)) = \pi(m)/2$.

Step 4 – Orbit length. The orbit length of (b_0, e_0) under $T = Q^2$ divides $\pi(m)/2$. A state is primitive if its orbit length equals $\pi(m)/2$; non-primitive states yield shorter orbits. □

Proposition 12.2 (Quadratic Norm Invariant and $\mathbb{Q}(\sqrt{5})$ Structure)**.** *The QA map T is the linear representation of multiplication by φ^2 in $\mathbb{Z}[\varphi]$. The quadratic form $f(b, e) = b^2 + be - e^2$ is the algebraic norm $N(b + e\varphi)$ in $\mathbb{Q}(\sqrt{5})$, and is invariant under T. The mod-m value of f classifies QA orbits.*

Proof. Step 1 – Algebraic identification. Multiplication by φ has matrix Q (since $\varphi(b + e\varphi) = b\varphi + e\varphi^2 = e + (b + e)\varphi$). Therefore $T = Q^2$ represents multiplication by φ^2.

Step 2 – Norm identification. $N(b+e\varphi) = (b+e\varphi)(b+e\bar{\varphi}) = b^2 + be(\varphi+\bar{\varphi}) + e^2\varphi\bar{\varphi} = b^2 + be - e^2 = f(b,e)$, using $\varphi+\bar{\varphi} = 1$ and $\varphi\bar{\varphi} = -1$.

Step 3 – Invariance. Since $N(\varphi^2) = N(\varphi)^2 = (-1)^2 = 1$, multiplication by φ^2 preserves the norm: $f(T(b,e)) = f(b,e)$. □

Geometric identity. $\det\left(\begin{smallmatrix} b & d \\ e & a \end{smallmatrix}\right) = ba - ed = b(b+2e) - e(b+e) = b^2 + be - e^2 = f(b,e)$. Thus $f(b,e)$ is the determinant of the matrix whose columns are the state before and after one QA step.

Corollary 12.3 (Orbit classification, $m = 9$)**.** *The orbit type of (b_0, e_0) is determined by the 3-adic valuation of $N(b_0 + e_0\varphi)$:*

$v_3(f(b_0,e_0))$	*Orbit length*	*Orbit type*
≥ 4	*1*	*Singularity* $(9,9)$
2 *or* 3	*4*	*Satellite*
0	*12*	*Cosmos (primitive)*

Verified for all 81 states of $\{1,\ldots,9\}^2$.

Fibonacci projective flow. The projective ratio $z_t = e_t/(b_t+e_t)$ evolves as $z_{t+1} = (1+z_t)/(2+z_t)$ under T, with unique fixed point $z^* = (\sqrt{5}-1)/2 = 1/\varphi \approx 0.618$.

H_{QA} decomposition. Setting $d = b+e$ and $z = e/d$:

$$H_{\text{raw}} = \frac{1-z^2}{4(1+z^2)} + \frac{e}{8}.$$

The first term equals $\cos(2\theta)/4$ where $\theta = \arctan(z)$; the second, $e/8 = \sigma_{od}/4$, follows from the Fibonacci identity $a+b = 2d$. Writing $M = \left(\begin{smallmatrix} b & e \\ d & a \end{smallmatrix}\right)$:

$$4H_{\text{raw}} = \sigma_d(M) + \sigma_{od}(M), \quad \sigma_d = \frac{M_{11}M_{22}}{M_{12}^2 + M_{21}^2} \text{ (degree-0)}, \quad \sigma_{od} = \frac{M_{12}M_{21}}{M_{11}+M_{22}} \text{ (degree-1)}.$$

The formula satisfies transpose symmetry $\sigma(M) = \sigma(M^\top)$ and bilinear identity $4H_{\text{raw}}(e^2+d^2)(b+a) = ba(b+a) + ed(e^2+d^2)$ (verified for all seven experimental substrates). Defining $\Lambda = \operatorname{diag}(\sigma_d, \sigma_{od})$: $4H_{\text{raw}} = \operatorname{tr}(\Lambda)$ and $2H_{\text{raw}} = R(\Lambda, (1,1)^\top/\sqrt{2})$ (the Rayleigh quotient at the equal-weight vector; see Proposition 12.4 for why $(1,1)^\top/\sqrt{2}$ is canonical). The factor $1/4 = \operatorname{tr}(\Lambda)/(2n)|_{n=2}$ is the normalised-trace scaling of the 2×2 operator.

Proposition 12.4 (Equal-Weight Rayleigh Vector)**.** *The identity $2H_{\text{raw}} = R(\Lambda, v_0)$ with $v_0 = (1,1)^\top/\sqrt{2}$ is the unique unit-vector Rayleigh representation that simultaneously satisfies:*

(i) ***Transpose symmetry.*** *σ_d and σ_{od} are individually invariant under $M \mapsto M^\top$ (rows and columns of the QA step matrix exchanged), so Λ is transpose-symmetric; v_0 is the eigenvector of the symmetry group $\mathbb{Z}/2$ acting on $\mathbb{R}^2$ by coordinate swap.*

(ii) ***Before/after symmetry.*** *$M = \left(\begin{smallmatrix} b & e \\ d & a \end{smallmatrix}\right)$ has two natural pairs: (b,e) (state before T) and (d,a) (state after T). Neither is algebraically preferred, since T is a bijection on every orbit. Equal weighting of the two diagonal entries σ_d, σ_{od} of Λ corresponds exactly to v_0.*

(iii) ***Trace consistency.*** *$4H_{\text{raw}} = \text{tr}(\Lambda) = 2\,R(\Lambda, v_0)$, so v_0 recovers the normalised-trace formula with the unique scalar $1/\sqrt{2}$ normalization.*

Equivalently, $v_0 = (1,1)^\top/\sqrt{2}$ is the unique unit vector in the positive orthant fixed by the swap $v_1 \leftrightarrow v_2$.

Theorem 12.5 (Orbit Contraction)**.** *Let $\mathcal{O} = \{s_0, \ldots, s_{L-1}\}$ be a finite QA orbit of length $L = \pi(m)/2$ with per-step effective rate $\eta_{\text{eff}}^{(t)} = \text{lr} \cdot \text{gain} \cdot H_{\text{QA}}^{(t)}$. Define $\kappa_{\min}(\mathcal{O}) = \min_t(1 - |1 - \eta_{\text{eff}}^{(t)}|)$. If $\kappa_{\min}(\mathcal{O}) > 0$, then $0 < \eta_{\text{eff}}^{(t)} < 2$ for every $t \in \{0, \ldots, L-1\}$.*

Proof. $\kappa_{\min} > 0 \Leftrightarrow |1 - \eta_{\text{eff}}^{(t)}| < 1 \Leftrightarrow 0 < \eta_{\text{eff}}^{(t)} < 2$ for all t. Since the orbit is finite and enumerable, this condition can be checked exactly. □

References

[1] A. Vaswani et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

[2] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu. A comprehensive survey on graph neural networks. *IEEE Trans. Neural Netw. Learn. Syst.*, 32(1):4–24, 2020.

[3] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv:1412.6980*, 2014.

[4] L. Bottou. Stochastic gradient descent tricks. In *Neural Networks: Tricks of the Trade*, pp. 421–436. Springer, 2012.

[5] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. *arXiv:1609.02907*, 2016.

[6] S. Ruder. An overview of gradient descent optimization algorithms. *arXiv:1609.04747*, 2016.

[7] D. Silver et al. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016.

[8] T. Wolf et al. HuggingFace's Transformers: State-of-the-art natural language processing. *arXiv:1910.03771*, 2019.

[9] D. D. Wall. Fibonacci primitive roots and the period of the Fibonacci sequence modulo a prime. *Am. Math. Monthly*, 67(6):525–532, 1960.